Vivek Tiwari

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https://vivtiw.github.io/

PROFESSIONAL SUMMARY

Senior/Staff Backend Software Engineer with 10+ years of experience shipping scalable, secure systems across cloud and on-prem environments. Track record of delivering full-stack backend features end-to-end at Fortinet, Motorola Solutions, Zscaler, and Oracle — across security, healthcare, and climate tech. AI-native builder who uses Cursor and Claude to ship faster, automate workflows, and build SaaS products with lean teams.

EDUCATION

University of Victoria Victoria, BC, Canada
MASc, Electrical and Computer Engineering.
National Institute of Technology, Durgapur West Bengal, India
B.Tech in Electronics and Communication Engineering.

EXPERIENCE

Principal Member of Technical Staff May 2024 – Present
Oracle *Remote, CA, USA*

- Designed and delivered SMART-on-FHIR v2 authorization flows for third-party healthcare apps to securely launch against OHIN APIs, laying the groundwork for broader EHR and partner integrations.
- Led QHIN conformance testing for Oracle Health Information Network (OHIN), passing all Sequoia Project RCE certification suites and validating every XUA transaction end-to-end with partner networks.
- Accelerated feature development using Oracle Code Assist, Cline, and Codex CLI to automate boilerplate generation and increase code robustness across Python, Java, REST API, and CI/CD workflows.
- Refined end-to-end API compliance testing frameworks for FHIR and TEFCA standards, collaborating with multiple internal teams to streamline interoperability validation and reduce deployment timelines.

Sr Software Developer February 2024 – May 2024
Zscaler *Vancouver, BC, Canada*

- Improved DLP detection reliability by identifying coverage gaps, refining internal detection logic, and building automation to ensure consistent daily rule execution and eliminate missed detections.

Sr Software Developer, Climate Tech/Sustainability November 2021 – January 2024
BMO Radicle *Remote, Canada*

- Cut software release time by 60% (from over 5 hours to under 2 hours) by migrating the stack from Ruby on Rails to Python/Django and rebuilding the CI/CD pipeline from scratch on GitHub Actions.
- Built a linear regression model using TensorFlow and Python for carbon footprint estimation and energy consumption optimization, serving customers and climate-smart advisors.
- Integrated third-party energy data providers via webhook APIs using Ruby on Rails to automate aggregation of energy consumption data, simplifying data collection for climate-smart advisors.

Sr Software Developer, Access Control September 2016 – October 2021
Motorola Solutions *Vancouver, BC, Canada*

- Delivered critical security features for Access Control Manager (ACM), including JWT-based authentication/authorization, rate limiting via iptables, and SSL certificate management with OpenSSL X.509, reducing attack surface and preventing potential revenue loss.
- Reduced report generation latency by 75% (from approximately 20 minutes to under 5 minutes) using Redis caching, batched APIs, and asynchronous processing.
- Implemented real-time biometric integration (Virdi) via WebSockets in a Ruby on Rails system, enabling low-latency device communication and contributing to approximately 20% increase in adoption and revenue.
- Engineered hardware-integrated test automation using Raspberry Pi to simulate card-swiping workflows in RSpec, eliminating approximately 2 days of manual testing effort per engineer and improving system test coverage.

Software Developer, Sandbox/Data May 2014 – August 2016
Fortinet *Vancouver, BC, Canada*

- Architected and deployed a highly available Web Filtering application on AWS using Elastic Beanstalk, pioneering the team's adoption of AWS as a SaaS platform and enabling performance quantification of the sniffer engine.
- Built a Test Automation framework in Python using Scapy and Pytest to analyze and debug TCP/IP and HTTP traffic, improving isniff engine performance to approximately 90% accuracy.

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification, Coursera, 2023
- Advanced Learning Algorithms, Coursera, 2023

SKILLS

Backend & Systems: Ruby (Rails), Python, Java, TypeScript, Perl, PHP, REST APIs, Microservices
Cloud & DevOps: AWS, OCI, Docker, Kubernetes, CI/CD, GitHub Actions, Bash/PowerShell, Git
Data & Observability: ELK, Grafana, Fluent Bit, Redis
Security & AI: OWASP, AuthN/AuthZ, TensorFlow, Cursor/Claude, FHIR, TEFCA